

WEEK 03 FINAL PROJECT REPORT

Password Security & Password Cracking

W3-PM1	Password Cracking with JTR
W3-PM2	Password Cracking with NW Tools
Environment	Kali Linux • VMware Workstation
Assessment	Authorized Educational Laboratory

Practical documentation of Week 03 activities

This report documents the methodology, evidence, results, observations and security lessons from the assigned password-security laboratory.

Confidentiality Note

The recovered passwords belong only to intentionally provided training files. The techniques described are for authorized educational use, security labs and controlled testing.

1. Executive Summary

During Week 03 of the NetworkWalks Cybersecurity Internship (Batch B082), I completed two practical modules focused on password security and password recovery: W3-PM1, Password Cracking with John the Ripper (JTR), and W3-PM2, Password Cracking with NW Tools.

For W3-PM1, I worked with three password-protected PDF files. The password material was converted into crackable PDF hash representations and saved as **hash1.txt**, **hash2.txt** and **hash3.txt**. Instead of using the graphical Johnny interface shown in the task material, I used John the Ripper directly from the Kali Linux command line. The exercise successfully recovered the password for all three laboratory PDFs. PDF 1 and PDF 2 used the same password, while PDF 3 used a different password.

For W3-PM2, I used the NetworkWalks Hash Calculator and dictionary-based password-cracking workflow. The extracted PDF hash was supplied to the tool, candidate passwords were tested, and a successful password match was obtained. The result was then verified against the protected PDF.

The combined exercises demonstrated the relationship between password strength, password reuse, hash handling and offline password-recovery techniques. They also reinforced the importance of strong, unique passwords and careful handling of password hashes.

2. Objectives

- Understand the purpose of password-security testing.
- Understand how a protected PDF can be represented as a crackable password hash.
- Use John the Ripper on Kali Linux through the command line.
- Identify and use the appropriate PDF hash format.
- Perform controlled dictionary-based password recovery.
- Use NetworkWalks tools for hash processing and password testing.
- Verify recovered passwords by opening the corresponding protected PDF.
- Document technical evidence and results in a professional manner.
- Understand the security impact of password reuse and predictable passwords.

3. Scope and Ethical Considerations

All work documented in this report was performed against intentionally supplied training files as part of the assigned NetworkWalks cybersecurity internship laboratory. The activity was limited to password-protected PDF files provided for the exercise.

Authorized use: The techniques described should only be used against files, accounts or systems for which explicit permission has been provided. Password recovery against third-party material without authorization is outside the scope of this report.

The Week 03 task material explains that John the Ripper is used by security professionals to test password strength and supports password-protected formats including PDF. It also states that Kali Linux users can run JTR directly because it is available in Kali Linux. The supplied task demonstrates a graphical Johnny workflow, while this implementation used the command-line JTR approach.

4. Laboratory Environment

Component	Configuration
Host / Virtualization	VMware Workstation
Guest Operating System	Kali Linux
Primary Tool	John the Ripper (JTR)
Execution Method	Command Line
File Type	Password-protected PDF
Password Recovery	Dictionary-based testing
Secondary Tools	NetworkWalks Hash Calculator / Password Cracker
Evidence	Terminal screenshots, browser screenshots and protected-PDF verification

5. Tools and Their Purpose

Tool	Purpose in the Assessment
John the Ripper	Process PDF password hashes and perform password recovery.
Kali Linux	Security testing platform used to execute JTR from the terminal.
NetworkWalks Hash Calculator	Extract/process a crackable representation of a protected PDF.
NetworkWalks Password Cracker	Perform dictionary-based candidate password testing.
PDF Viewer	Verify recovered passwords against the protected files.

6. Methodology

The overall workflow followed the sequence below:

Protected PDF → Hash Extraction → Save Hash File → Identify PDF Format → Password Testing → Password Recovery → PDF Verification → Evidence Collection

The supplied JTR module instructs the learner to copy the extracted hash, remove unwanted extra characters when necessary, and save the hash in a text file beginning with the PDF hash format. The module then uses the saved hash file for the JTR workflow.

7. W3-PM1 — Password Cracking with JTR

The assigned module introduces John the Ripper and Johnny, the graphical interface for JTR. The task is to recover the password of a protected PDF. On Kali Linux, I used the command-line implementation of JTR instead of the Windows/Johnny graphical workflow.

7.1 PDF 1

Input: **My Locked PDF1.pdf**

Hash file: **hash1.txt**

Command used:

```
john --format=PDF hash1.txt
```

John the Ripper loaded the PDF hash and processed candidate passwords from its configured wordlist. The password was successfully recovered and subsequently verified by opening the protected PDF.

7.2 PDF 2

Input: **My Locked PDF2.pdf**

Hash file: **hash2.txt**

Command used:

```
john --format=PDF hash2.txt
```

PDF 2 was successfully processed and its password was recovered. The recovered password matched the password recovered for PDF 1, demonstrating password reuse across two protected files.

7.3 PDF 3

Input: **My Locked PDF3.pdf**

Hash file: **hash3.txt**

Command used:

```
john --format=PDF hash3.txt
```

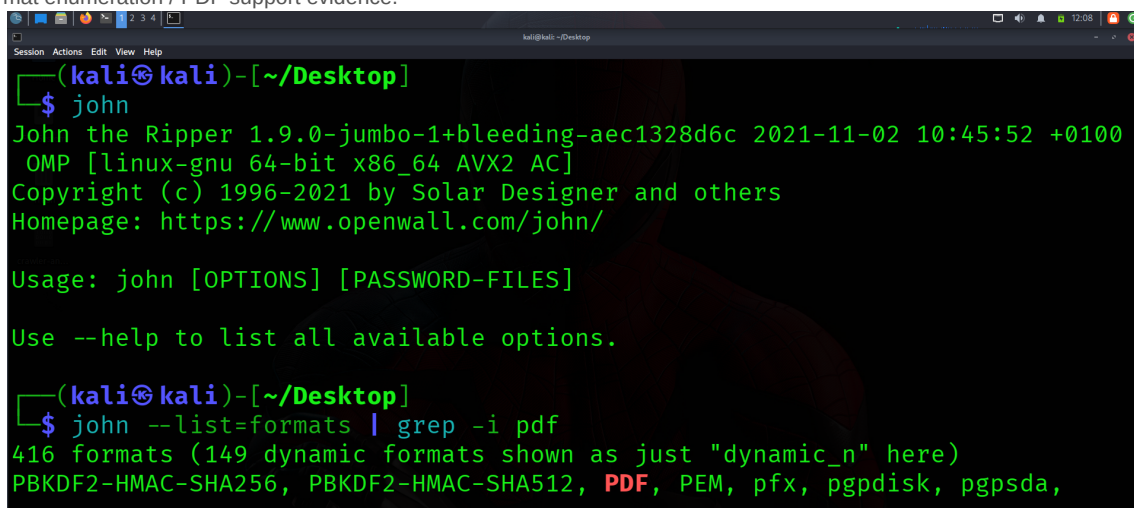
PDF 3 was successfully processed and its password was recovered. Verification showed that PDF 3 used a different password from PDF 1 and PDF 2.

7.4 W3-PM1 Result Summary

File	Hash File	JTR Result	Password Relationship
PDF 1	hash1.txt	Recovered	Shared with PDF 2
PDF 2	hash2.txt	Recovered	Shared with PDF 1
PDF 3	hash3.txt	Recovered	Different from PDF 1/2

7.5 JTR Terminal Evidence

JTR format enumeration / PDF support evidence.



```
(kali@kali)-[~/Desktop]
$ john
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100
OMP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.

(kali@kali)-[~/Desktop]
$ john --list=formats | grep -i pdf
416 formats (149 dynamic formats shown as just "dynamic_n" here)
PBKDF2-HMAC-SHA256, PBKDF2-HMAC-SHA512, PDF, PEM, pfx, pgpdisk, pgpsda,
```

JTR execution and successful password recovery evidence.

```
(kali㉿kali)-[~/Desktop]
└─$ john --format=PDF pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password1 (?)
1g 0:00:00:00 DONE 2/3 (2026-08-24 12:10) 100.0g/s 12800p/s 12800c/s 12800C
/s 123456..john
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.
```

Additional JTR execution/result evidence.

```
(kali㉿kali)-[~/Desktop]
└─$ john --format=PDF pdf_hash2.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password1 (?)
1g 0:00:00:00 DONE 2/3 (2026-08-25 10:29) 50.00g/s 6400p/s 6400c/s 6400C/s 123456..john
n
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.
```

8. W3-PM2 — Password Cracking with NW Tools

The second module used NetworkWalks security tools to demonstrate a browser-based password-security workflow. The practical process consisted of extracting a crackable PDF hash, supplying that hash to the password-cracking interface, running a dictionary attack and verifying the resulting password against the protected PDF.

8.1 Hash Extraction

The protected PDF was processed using the NetworkWalks Hash Calculator. The tool displayed a PDF hash in a format suitable for password-security testing. The resulting value was then used as input to the password-cracking exercise.

8.2 Dictionary-Based Password Testing

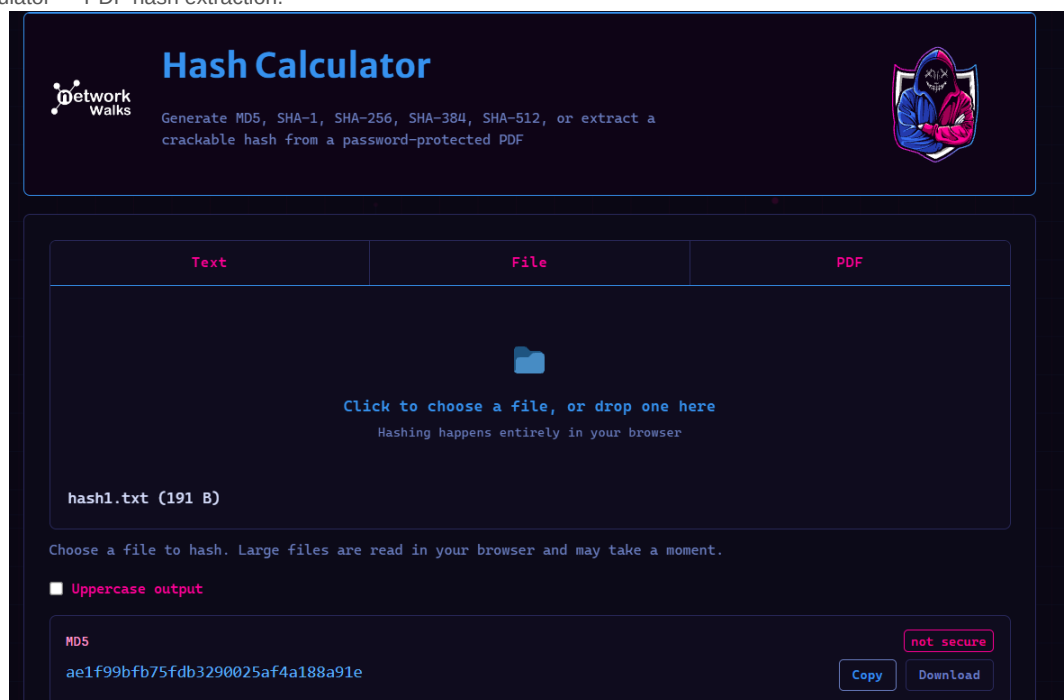
The extracted PDF hash was entered into the NetworkWalks password-cracker interface. A built-in wordlist was used to test candidate passwords. The interface showed candidate attempts and reported a successful match.

8.3 Verification

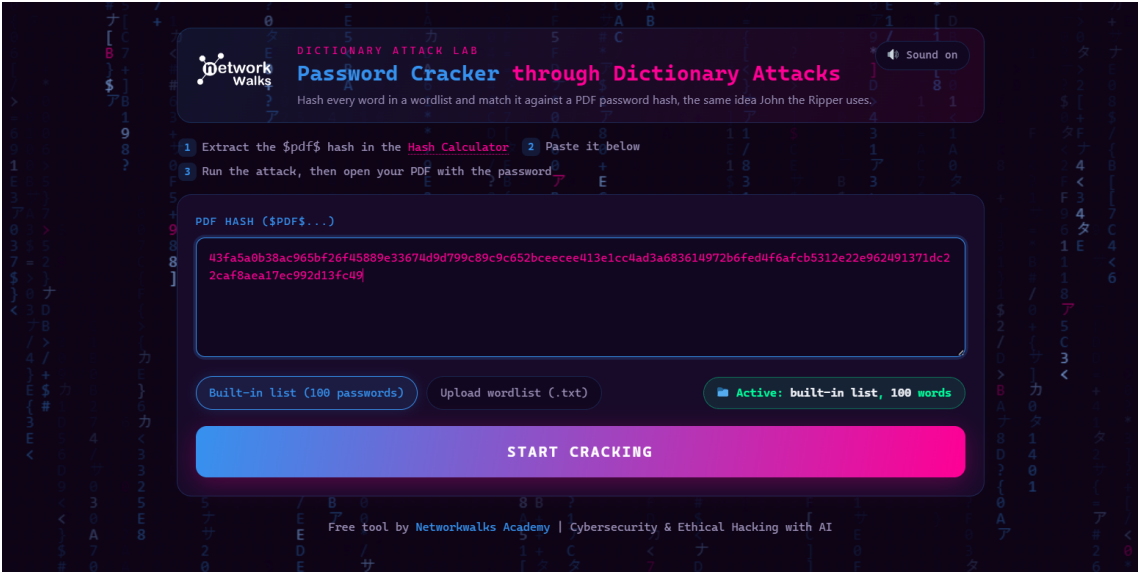
The recovered password was used to open the protected training PDF, confirming that the candidate matched the PDF password.

8.4 NetworkWalks Tool Evidence

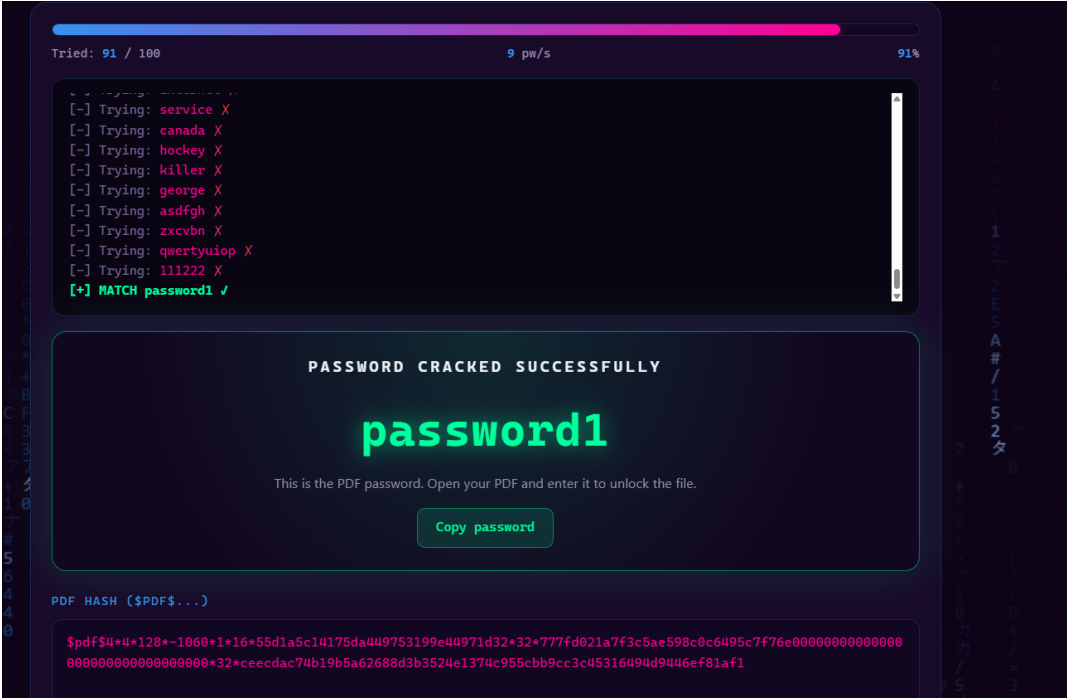
Hash Calculator — PDF hash extraction.



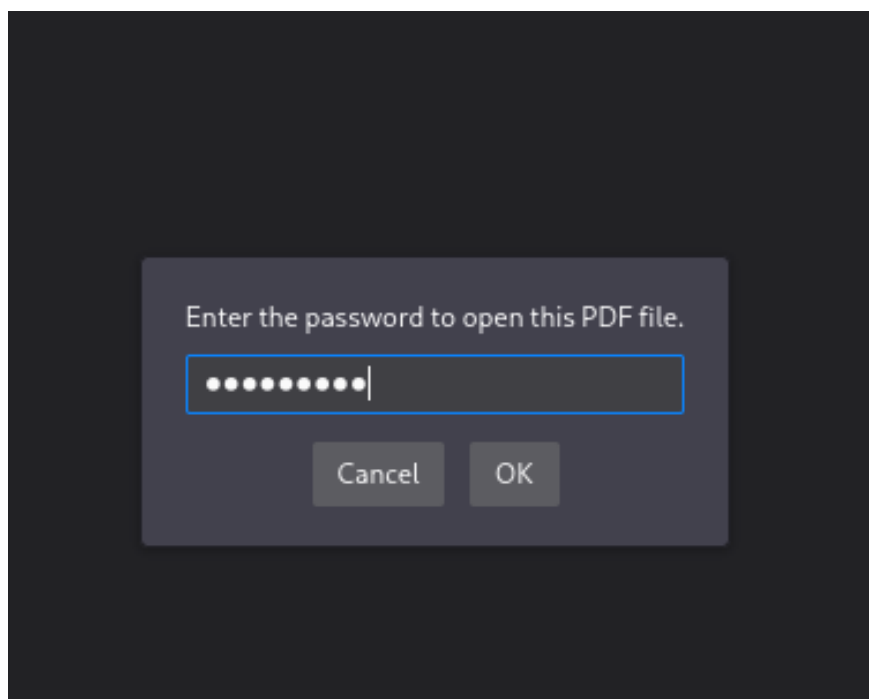
Password Cracker — hash supplied for dictionary testing.



Password Cracker — successful password match.



Protected PDF — password verification evidence.



9. Overall Results

Activity	Result
W3-PM1 — PDF 1	Password successfully recovered
W3-PM1 — PDF 2	Password successfully recovered; same as PDF 1
W3-PM1 — PDF 3	Password successfully recovered; different from PDF 1/2
W3-PM2 — Hash extraction	Completed
W3-PM2 — Dictionary testing	Successful password match
PDF verification	Completed
Evidence collection	Completed
Documentation	Completed

10. Security Observations

Password reuse: PDF 1 and PDF 2 used the same password. If a password is reused across multiple resources, recovery of that password can affect more than one protected resource.

Predictable passwords: Dictionary-based testing can recover passwords when they appear in a candidate list. This demonstrates why simple or commonly used passwords should not be used to protect sensitive files.

Hash sensitivity: Extracted password hashes should be handled as sensitive security material because they may be subjected to offline password-recovery attempts.

Verification matters: A password-recovery result should be verified against the original protected file rather than relying only on a tool's displayed output.

11. Security Recommendations

- Use long, complex and unpredictable passwords for protected documents.
- Avoid reusing the same password across multiple files or services.
- Use unique passwords for sensitive documents and accounts.
- Protect password hashes and avoid exposing them unnecessarily.
- Where appropriate, use modern encryption and secure document-protection mechanisms.
- Use password managers to generate and store unique passwords.
- Perform password-strength assessments only within an authorized scope.
- Remove training or test passwords and hashes from production environments after exercises are completed.

12. Learning Outcomes

This project strengthened practical skills in Kali Linux, Linux command-line operation, John the Ripper, PDF hash handling, dictionary-based password recovery, browser-based security tools, evidence collection and technical reporting.

- Understanding the workflow from protected file to password hash.
- Selecting and specifying the correct JTR format.
- Executing JTR from the Kali Linux terminal.
- Reading and interpreting JTR output.
- Understanding dictionary-based password testing.
- Recognizing the security impact of password reuse.
- Verifying results using the original protected document.
- Maintaining an evidence-based cybersecurity report.

13. Conclusion

Week 03 provided practical exposure to password-security assessment and controlled password-recovery techniques. The two project modules demonstrated both a command-line workflow using John the Ripper on Kali Linux and a graphical workflow using NetworkWalks security tools.

The JTR exercise successfully recovered the passwords of all three training PDFs. PDF 1 and PDF 2 were found to share the same password, while PDF 3 used a different password. The NetworkWalks tool workflow also successfully demonstrated PDF hash extraction and dictionary-based password recovery.

The most important security lesson from the exercise is that password protection is only as strong as the password used. Strong, unique and unpredictable passwords reduce the effectiveness of dictionary-based password-recovery attempts and reduce the impact of password reuse.

14. Evidence Register

Evidence	Purpose
hash1.txt	PDF 1 password hash used with JTR
hash2.txt	PDF 2 password hash used with JTR
hash3.txt	PDF 3 password hash used with JTR

Evidence	Purpose
JTR screenshots	Terminal commands, hash loading and recovery output
Hash Calculator screenshot	NetworkWalks PDF hash extraction evidence
Password Cracker screenshots	Dictionary testing and successful match
Protected PDF screenshot	Password verification evidence

15. References / Source Material

1. NetworkWalks Cybersecurity & Ethical Hacking Project Tasks — Week 3, Project Module 1: Password Cracking with JTR. The module describes JTR/Johnny, the protected-PDF exercise and the Kali Linux command-line option.
2. Practical evidence collected during the Week 03 laboratory exercise, including terminal output, NetworkWalks tool screenshots and PDF verification screenshots.